Practice paper 3 – Exercise 1

Integration and L^p spaces | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. For $n \geq 1$, let $f_n(x) = \mathbf{1}_{(0,1/n)}(x)$ on $(0,1)$, equipped with Lebesgue measure. Thus f_n equals 1 on $(0,1/n)$ and 0 elsewhere.

(a) **[2 points]** Fix $x > 0$ and explain why $f_n(x) = 0$ for all sufficiently large n. Find the pointwise limit.

(b) **[3 points]** Compute $\int_0^1 f_n \, dx$, $\|f_n\|_{L^1}$, and $\|f_n\|_{L^2}$. Determine whether $f_n \to 0$ in L^1 and in L^2. *Hint:* $f_n^2 = f_n$ and the interval $(0,1/n)$ has length $1/n$.

(c) **[2 points]** Use the bound $0 \leq f_n \leq 1$ to apply the dominated convergence theorem. State the hypotheses you use.

Practice paper 3 – Solution 1

Detailed worked solution

(a) Pointwise limit. Let $x \in (0,1)$ be fixed. Take an integer $N > 1/x$. Then, whenever $n \geq N$, $1/n < x$, so x is outside the interval $(0, 1/n)$. By the definition of the indicator function, $f_n(x) = 0$. Consequently,

$$\boxed{f_n(x) \to 0 \quad \text{for every } x \in (0,1).}$$

The value of N may depend on x; this is allowed in pointwise convergence.

(b) Integral and norms. The integral of an indicator equals the measure of its set:

$$\int_0^1 f_n(x)\,dx = \lambda((0,1/n)) = \frac{1}{n}.$$

Since f_n is nonnegative and takes only values zero and one, $|f_n| = f_n$ and $f_n^2 = f_n$. Therefore

$$\|f_n\|_{L^1} = \frac{1}{n}, \qquad \|f_n\|_{L^2} = \left(\int_0^1 f_n(x)^2\,dx\right)^{1/2} = \sqrt{\frac{1}{n}} = \frac{1}{\sqrt{n}}.$$

Both quantities tend to zero, so

$$\boxed{f_n \to 0 \text{ in } L^1(0,1) \text{ and in } L^2(0,1).}$$

Although the height of the function stays equal to one, the interval on which it is nonzero becomes shorter.

(c) Dominated convergence. Every interval $(0, 1/n)$ is Borel measurable, so each indicator f_n is measurable. Part (a) proves pointwise convergence to zero. Furthermore,

$$|f_n(x)| \leq g(x) := 1 \quad \text{for every } n, x, \qquad \int_0^1 g(x)\,dx = 1 < \infty.$$

The bound is independent of n and is integrable. The dominated convergence theorem applies and gives

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx = 0.$$

This agrees with the explicit integral $1/n$.

Key point. For an indicator, the integral is the measure of the underlying set; the L^2 norm is the square root of that measure.

Practice paper 3 – Exercise 2

Probability and conditioning | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. An urn is selected at random: urn A with probability $1/2$ and urn B with probability $1/2$. A single ball is then drawn. The probability of drawing a red ball is $3/4$ from urn A and $1/4$ from urn B.

Let $X = 0$ if urn A is selected and $X = 1$ if urn B is selected. Let $Y = 1$ if the ball is red and $Y = 0$ otherwise.

(a) **[2 points]** Find $\mathbb{P}(Y = 1)$ using the law of total probability. What is $\mathbb{E}[Y]$?

(b) **[3 points]** Compute $\mathbb{P}(X = 0 \mid Y = 1)$ using Bayes' formula. Are X and Y independent? Compare $\mathbb{P}(Y = 1 \mid X = 0)$ with $\mathbb{P}(Y = 1)$.

(c) **[2 points]** Find $\mathbb{E}[Y \mid X = 0]$ and $\mathbb{E}[Y \mid X = 1]$. Verify the tower property by averaging these two values with the probabilities of selecting the urns.

Practice paper 3 – Solution 2

Detailed worked solution

(a) Total probability. The cases $X = 0$ and $X = 1$ partition the sample space. Therefore

$$\begin{aligned}\mathbb{P}(Y = 1) &= \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0) \\ &\quad + \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1) \\ &= \frac{3}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{3}{8} + \frac{1}{8} = \boxed{\frac{1}{2}}.\end{aligned}$$

Because Y takes values zero and one,

$$\mathbb{E}[Y] = 0\mathbb{P}(Y = 0) + 1\mathbb{P}(Y = 1) = \frac{1}{2}.$$

(b) Bayes' formula and independence. First calculate the relevant joint probability:

$$\mathbb{P}(X = 0, Y = 1) = \mathbb{P}(Y = 1 \mid X = 0)\mathbb{P}(X = 0) = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}.$$

Since $\mathbb{P}(Y = 1) = 1/2 > 0$, Bayes' formula gives

$$\mathbb{P}(X = 0 \mid Y = 1) = \frac{\mathbb{P}(X = 0, Y = 1)}{\mathbb{P}(Y = 1)} = \frac{3/8}{1/2} = \boxed{\frac{3}{4}}.$$

Observing a red ball makes urn A more likely: its probability rises from $1/2$ to $3/4$. Also,

$$\mathbb{P}(Y = 1 \mid X = 0) = \frac{3}{4} \neq \frac{1}{2} = \mathbb{P}(Y = 1).$$

Thus knowing the urn changes the probability of red, and X, Y are not independent.

(c) Conditional expectation and the tower property. For a zero–one variable, the conditional mean is its conditional probability of being one. Hence

$$\mathbb{E}[Y \mid X = 0] = \frac{3}{4}, \qquad \mathbb{E}[Y \mid X = 1] = \frac{1}{4}.$$

Averaging over the possible values of X yields

$$\mathbb{E}[\mathbb{E}[Y \mid X]] = \frac{3}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2} = \mathbb{E}[Y].$$

Key point. Bayes' formula reverses the direction of conditioning: the probability of red given an urn differs from the probability of that urn given red.